195830: metamonzogranite, Mount Seabrook

(Moorarie Supersuite, Burringurrah Domain, Gascoyne Province, Capricorn Orogen)

Location and sampling

Robinson Range (SG 50-7), Moorarie (2446)
MGA Zone 50, 573504E 7167051N

Sampled on 10 May 2011

This sample was collected from an outcrop in a creek on Trillbar Station, about 4.5 km south-southwest of Trillbar homestead, 1.5 km south-southeast of Mount Seabrook Talc Mine, and 1.3 km east of Mount Seabrook.

Geological context

The unit sampled is a metamonzogranite assigned to the 1817–1773 Ma Moorarie Supersuite of the Burringurrah Domain (Occhipinti and Meyers, 1999; Sheppard et al., 2010). The 1945–1620 Ma Burringurrah Domain includes all igneous, metasedimentary and metamorphic rock units within the Gascoyne Province that are younger than the 2555–1945 Ma Glenburgh Terrane and not included within, and mostly older than, the 1673–1455 Ma Edmund Group. The Moorarie Supersuite is dominated by granitic rocks of granodioritic to monzogranitic composition, and was intruded into the Gascoyne Province, the adjacent Yarlarweelor Gneiss Complex, and the Ashburton Basin during the Capricorn Orogeny (Sheppard et al., 2010). The metamonzogranite at this locality is not foliated, and cross-cuts the main S_1 fabric in psammitic schist, a sample of which, collected about 0.6 km to the west-northwest, yielded a monazite age for metamorphism of 1991 ± 6 Ma (GSWA 195828, Fielding et al., 2022). Locally the monzogranite contains a weak northwest-trending foliation subparallel to F_2 axial planes in the schist farther east, suggesting that the monzogranite intruded after D_1 and prior to D_2. The Archean dates obtained for zircons from this sample (see below) indicate that either the zircons are inherited from an Archean rock, the sample represents an inclusion of Archean rock in Moorarie monzogranite, or the assignment of the unit is incorrect. Metamonzogranite of the Moorarie Supersuite, sampled about 7.6 km to the north-northeast, yielded an igneous crystallization age of 1808 ± 6 Ma (GSWA 142851, Nelson, 1998).

Petrographic description

The sample is a weathered, leucocratic metamonzogranite (Fig. 1) that contains 5–10% muscovite and biotite. The rock is mostly medium grained and equigranular, but is locally coarse grained to pegmatitic. A detailed petrographic description is not available for this sample.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, and anhedral to subhedral. The crystals are up to 120 µm long, and equant to elongate, with aspect ratios up to 4:1. In backscattered electron (BSE) images, some zircons exhibit concentric zoning, but most are internally homogeneous. BSE images of representative zircons are shown in Figure 2.

Analytical details

This sample was analysed on 8 September 2012, using SHRIMP-B. Six analyses of BR266 standard were obtained during the session, of which five analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.25% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.32%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Four analyses are >5% discordant. The dates obtained from these four analyses (Group D, Table 1) are unreliable, and are considered not to be geologically significant. The remaining 14 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios:

Group X comprises 13 analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3030–2653 Ma. The youngest 11 analyses yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2663 ± 4 Ma (MSWD = 0.73).

Group Z comprises one analysis (Table 1) which yields $^{207}Pb^*/^{260}Pb^*$ date of 1140 ± 21 Ma (1σ).

Figure 1. **Outcrop image for sample 195830: metamonzogranite, Mount Seabrook. The pen for scale marks the contact between metamonzogranite on the lower right of the image and psammitic schist on the upper left**

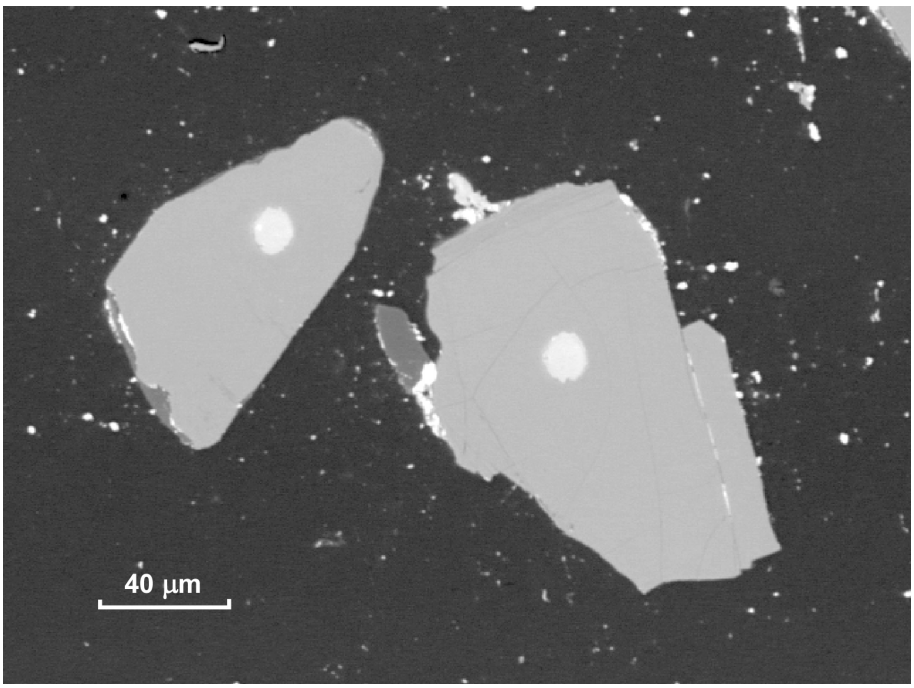

Figure 2. **BSE images of representative zircons from sample 195830: metamonzogranite, Mount Seabrook. Bright circular areas are SHRIMP analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 195830: metamonzogranite, Mount Seabrook

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$f204$ (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
X	8	8.1	446	114	0.26	0.038	1.948	0.027	0.18033	0.00077	1.949	0.027	0.17999	0.00078	2670	30	2653	7	-0.9
X	17	17.1	267	170	0.64	0.429	2.014	0.015	0.18440	0.00067	2.023	0.015	0.18056	0.00078	2589	17	2658	7	2.4
X	13	13.1	144	43	0.30	0.299	2.022	0.019	0.18327	0.00146	2.028	0.020	0.18059	0.00154	2584	21	2658	14	2.6
X	18	18.1	363	141	0.39	0.029	1.952	0.021	0.18089	0.00061	1.952	0.021	0.18063	0.00061	2666	24	2659	6	-0.5
X	12	12.1	515	124	0.24	-0.018	1.941	0.030	0.18049	0.00109	1.940	0.030	0.18065	0.00109	2680	34	2659	10	-1.0
X	1	1.1	158	81	0.51	0.013	1.986	0.017	0.18107	0.00080	1.987	0.017	0.18095	0.00080	2628	19	2662	7	1.1
X	5	5.1	251	100	0.40	0.028	1.870	0.025	0.18162	0.00067	1.871	0.025	0.18137	0.00068	2761	31	2665	6	-3.8
X	7	7.1	233	132	0.56	-0.014	1.911	0.015	0.18125	0.00070	1.910	0.015	0.18138	0.00071	2714	18	2665	6	-2.0
X	10	10.1	363	59	0.16	0.007	1.972	0.020	0.18179	0.00173	1.972	0.020	0.18173	0.00173	2644	22	2669	16	0.7
X	6	6.1	235	80	0.34	0.009	1.929	0.035	0.18190	0.00071	1.929	0.035	0.18182	0.00071	2692	40	2670	6	-1.0
X	16	16.1	394	61	0.15	0.053	2.012	0.031	0.18255	0.00062	2.013	0.031	0.18208	0.00063	2600	33	2672	6	2.5
X	11	11.1	128	60	0.47	-0.027	1.804	0.033	0.21300	0.00102	1.804	0.033	0.21324	0.00102	2843	43	2930	8	2.8
X	2	2.1	143	70	0.49	0.006	1.637	0.014	0.22688	0.00088	1.638	0.014	0.22683	0.00088	3073	22	3030	6	-1.6
Z	4	4.1	232	235	1.01	0.109	5.033	0.040	0.07865	0.00073	5.038	0.040	0.07773	0.00082	1167	9	1140	21	-3.0
D	9	9.1	209	260	1.24	1.170	16.815	0.352	0.05516	0.00224	17.014	0.359	0.04588	0.00316	368	8	-8	166	2365
D	3	3.1	54	67	1.23	0.719	4.792	0.058	0.07878	0.00120	4.827	0.060	0.07275	0.00219	1214	14	1007	61	-21.3
D	15	15.1	315	180	0.57	0.214	2.173	0.017	0.18298	0.00113	2.177	0.017	0.18107	0.00117	2436	16	2663	11	8.3
D	14	14.1	280	69	0.25	0.254	2.095	0.018	0.18356	0.00134	2.100	0.018	0.18128	0.00140	2510	18	2665	13	5.6

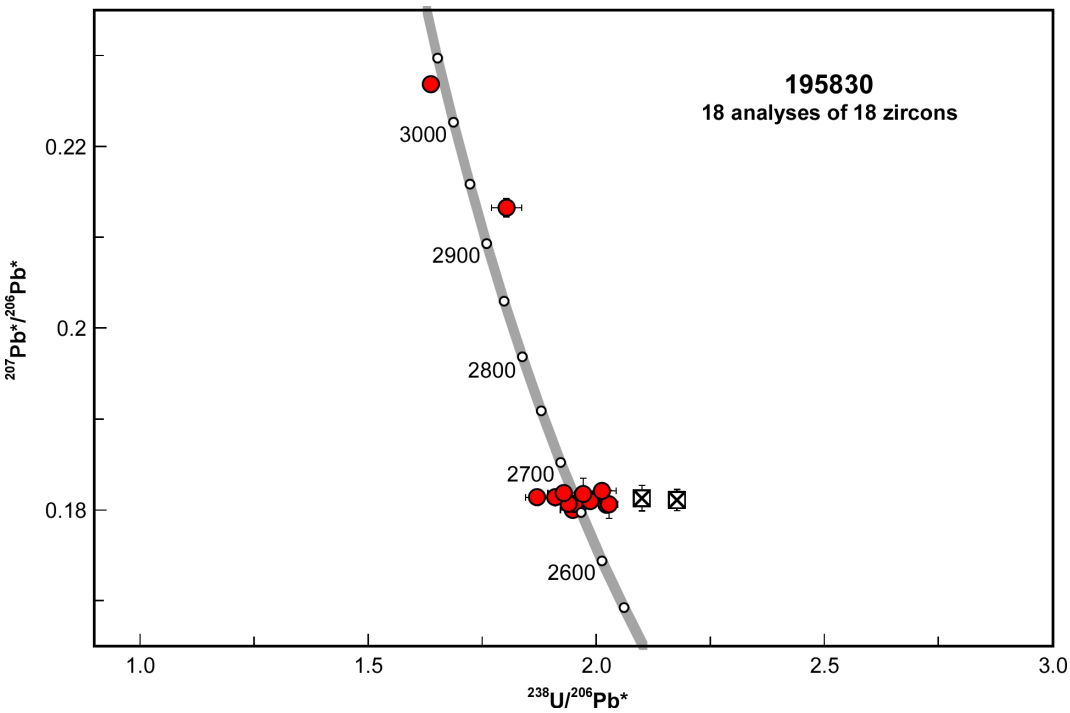

Figure 3. U–Pb analytical data for sample 195830: metamonzogranite, Mount Seabrook. Red circles indicate Group X (xenocrystic zircons); crossed squares indicate Group D (discordance >5%). Two analyses in Group D and one in Group Z are not shown

The dates of 3030–2653 Ma for the 13 analyses in Group X are interpreted as the ages of inherited zircons. The date of 2663 ± 4 Ma for 11 analyses in Group X is interpreted as the age of the dominant inherited component. The date of 1140 ± 21 Ma (1σ) for the single analysis in Group Z is interpreted as the age of a contaminant introduced during laboratory processing.

References

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Recommended reference for this publication

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